

Unit 11 - Exponential Functions

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The following B.E.S.T Standards will be covered in this unit:

MA.912.AR.5.3 Given a mathematical or real-world context, classify an exponential function as representing growth or decay.

- *Clarification 1: Within the Algebra 1 course, exponential functions are limited to the forms $f(x) = ab^x$, where b is a whole number greater than 1 or a unit fraction, or $f(x) = a(1 \pm r)^x$, where $0 < r < 1$.*

MA.912.AR.5.6 Given a table, equation or written description of an exponential function, graph that function and determine its key features.

- *Clarification 1: Key features are limited to domain; range; intercepts; intervals where the function is increasing, decreasing, positive or negative; constant percent rate of change; end behavior and asymptotes.*
- *Clarification 2: Instruction includes representing the domain and range with inequality notation, interval notation or set-builder notation.*
- *Clarification 3: Within the Algebra 1 course, notations for domain and range are limited to inequality and set builder.*
- *Clarification 4: Within the Algebra 1 course, exponential functions are limited to the forms $f(x) = ab^x$, where b is a whole number greater than 1 or a unit fraction or $f(x) = a(1 \pm r)^x$, where $0 < r < 1$.*

MA.912.AR.5.4 Write an exponential function to represent a relationship between two quantities from a graph, a written description or a table of values within a mathematical or real-world context.

- *Clarification 1: Within the Algebra 1 course, exponential functions are limited to the forms $f(x) = ab^x$, where b is a whole number greater than 1 or a unit fraction, or $f(x) = a(1 \pm r)^x$, where $0 < r < 1$.*
- *Clarification 2: Within the Algebra 1 course, tables are limited to having successive nonnegative integer inputs so that the function may be determined by finding ratios between successive outputs.*

Unit 11, Lesson 1: Exploring Exponential Growth Versus Decay



Benchmark

MA.912.AR.5.3 Given a mathematical or real-world context, classify exponential functions as representing growth or decay.



Learning Targets

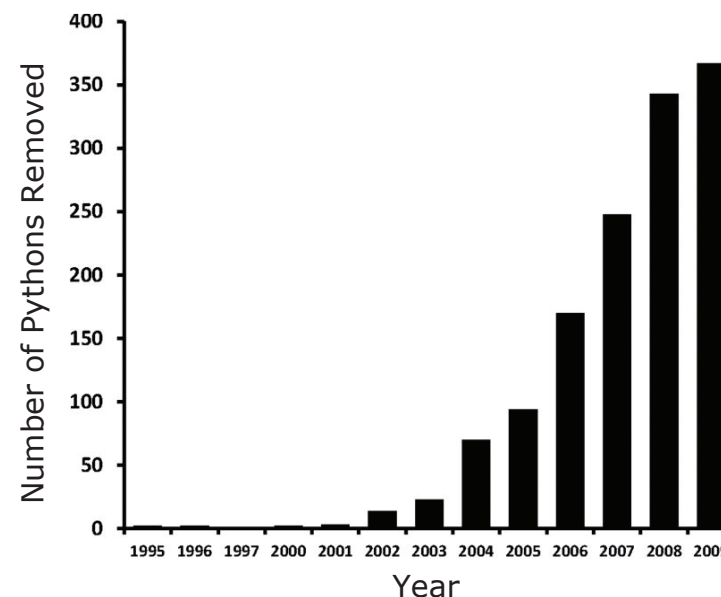
- ☐ I can classify a function as representing exponential growth or decay from a graph.
- ☐ I can classify a function as representing exponential growth or decay from a table.



11.1.1 Warm-Up: Invasive Snakes

1. The Burmese python is one of the largest species of snakes and is one of the most concerning invasive species in Everglades National Park. The graph below shows the number of pythons removed each year from 1995 to 2009.

Scientists now estimate the population of Burmese pythons in the Everglades is more than 100,000.



- a. What do you notice about the graph?
- b. Given the graph, are you surprised scientists now estimate more than 100,000 snakes in Everglades National Park?



11.1.2 Exploration: Dominoes and Stickers

1. Morris, a physics professor at the University of Toronto, set up a series of 13 dominoes, each roughly $1\frac{1}{2}$ times the size of the one knocking it over. The original domino is 5 millimeters (mm) tall and only 1 mm thick. The 13th domino is 3.2 feet (ft) tall and weighs about 100 pounds.



- a. What do you wonder about the dominoes?
- b. Do you think starting with the first domino, which is smaller than a Tic Tac, will result in the 13th domino falling?

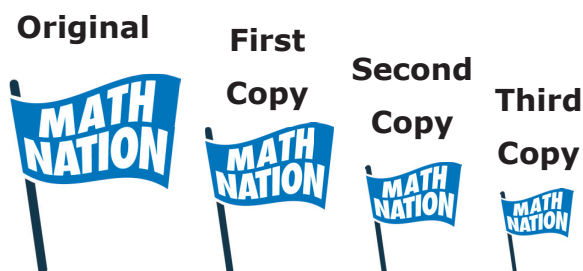
2. In a 1983 edition of the American Journal of Physics, Lorne Whitehead wrote about his discovery that as a domino falls it could topple things larger than itself. He found that a domino is capable of knocking down a domino that is 50% larger than itself.

- a. Use the information about the 13th domino to determine the height, in feet (ft), for the 14th and 15th dominos.

Domino	Height (ft)
13 th	3.2
14 th	
15 th	

- b. Discuss with your partner how the description that Dr. Morris uses, " $1\frac{1}{2}$ times itself," and the description Dr. Whitehead uses, "50% larger than itself," mean the same thing. Write your explanation or give an example to explain.

3. Ashley used a copy machine to make copies of a Math Nation sticker to observe how the size changes. Before making copies, she pressed the 75% zoom button.



- She made the first copy of the sticker.
- To make the second copy, she removed the original sticker from the copy machine and replaced it with the first copy of the sticker. She then pressed copy.
- To make the third copy, she removed the first copy of the sticker and replaced it with the second copy of the sticker. She then pressed copy.

How is each copy of the sticker related to the copy that was made before?



11.1.3 Collaboration: Graphs and Tables

There are two types of exponential functions: exponential growth and exponential decay.



An **exponential function** is a function with a constant percent rate of change.



Exponential growth describes an exponential function whose constant percent rate of change is greater than 100%.



Exponential decay describes an exponential function whose constant percent rate of change is greater than 0% but less than 100%.

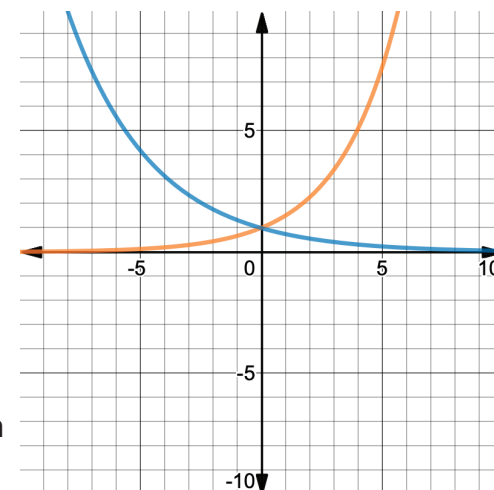
1. Use the descriptions of exponential growth and decay to determine what the orange graph represents and what the blue graph represents.

Orange:

- Exponential growth
- Exponential decay

Blue:

- Exponential growth
- Exponential decay



2. One of the graphs models the size of the dominoes, and the other models the size of the sticker. Discuss with your partner how to determine which graph, orange or blue, is the function that models the size of the dominoes. Write a summary of your discussion.

3. Each graph assumes the first sticker or the first domino is considered 100% of the original size.

What does the point $(0, 1)$ represent for the function that represents the dominoes?

4. The tables represent the two different contexts. Each table assumes that the first sticker or the first domino is 100% of the original size.

Table 1	
x	y
0	1
1	$\frac{3}{2}$
2	$\frac{9}{4}$
3	$\frac{27}{8}$
4	$\frac{81}{16}$
5	

Table 2	
x	y
0	1
1	$\frac{3}{4}$
2	$\frac{9}{16}$
3	$\frac{27}{64}$
4	$\frac{81}{256}$
5	

- a. In Table 1, find the y -value for $x = 5$ using the y -value for $x = 4$. Then, explain how you determined your answer.
- b. Elise found that in Table 1, when she divided $\frac{9}{4}$ by $\frac{3}{2}$ or $\frac{27}{8}$ by $\frac{9}{4}$, she got the same value. How does the value she found relate to the context?

- c. Use Elise's method and divide consecutive y -values in Table 2. How does the value relate to the context?

5. Discuss with your partner which table matches each context. Then, complete the statements based on your discussion.

Table 1 matches the ☐ stickers ☐ dominoes context because as

the x -values increase, the y -values in Table 1 are

☐ increasing ☐ decreasing at a constant percent rate

☐ between 0% and 100%.
☐ greater than 100%.

Table 2 matches the ☐ stickers ☐ dominoes context because as

the x -values increase, the y -values in Table 2 are

☐ increasing ☐ decreasing at a constant percent rate

☐ between 0% and 100%.
☐ greater than 100%.



11.1.4 Bring It Together: Exponential Equations



An exponential equation is a type of equation in which the input is an exponent. It can be written in the form $y = a(b)^x$.

The a -value of an exponential equation is the initial amount.

From the previous contexts with the first sticker and first domino, those were considered 100% or 1. However, this is not the case for every situation. For example, the **dimensions** of the sticker or the **height** of the first domino could have been used as the initial value.

The b -value of an exponential equation is the growth or decay factor.

Exponential Growth	Exponential Decay
b is the growth factor	b is the decay factor
$b = 1 + r$, where r is the growth rate	$b = 1 - r$, where r is the decay rate
$y = a(1 + r)^x$	$y = a(1 - r)^x$

1. Complete the statement.

When the equation is written in the form $y = a(b)^x$, the

b is greater than 1 when the equation is modeling

exponential ☐ decay ☐ growth, and is between 0 and 1 when

the equation is modeling exponential ☐ decay. ☐ growth.

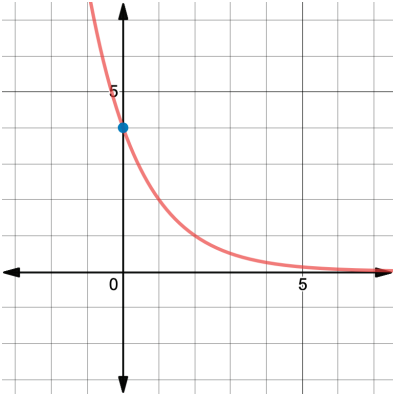
2. The equation that models the dominoes situation is $y = 1\left(1 + \frac{1}{2}\right)^x$ or $y = \left(\frac{3}{2}\right)^x$. The equation that models the sticker situation is $y = 1\left(1 - \frac{1}{4}\right)^x$ or $y = \left(\frac{3}{4}\right)^x$. Discuss with your partner how the value of b in each equation relates to the relationship observed in the tables.



11.1.5 Your Turn!

1. Determine if each representation of a function is an example of exponential decay or exponential growth. Justify your selection.

a.

Representation	Type of Exponential Function
	☐ Exponential decay ☐ Exponential growth
Justification	

b.

Representation	Type of Exponential Function
$y = 50(1 + 0.85)^x$	☐ Exponential decay ☐ Exponential growth
Justification	

c.

Representation	Type of Exponential Function												
<table border="1"> <thead> <tr> <th>x</th><th>y</th></tr> </thead> <tbody> <tr><td>-3</td><td>4</td></tr> <tr><td>-2</td><td>8</td></tr> <tr><td>-1</td><td>16</td></tr> <tr><td>0</td><td>32</td></tr> <tr><td>1</td><td>64</td></tr> </tbody> </table>	x	y	-3	4	-2	8	-1	16	0	32	1	64	○ Exponential decay ○ Exponential growth
x	y												
-3	4												
-2	8												
-1	16												
0	32												
1	64												
Justification													

d.

Representation	Type of Exponential Function
$y = 50(0.85)^x$	○ Exponential decay ○ Exponential growth
Justification	

e.

Representation	Type of Exponential Function												
<table border="1"> <thead> <tr> <th>x</th><th>y</th></tr> </thead> <tbody> <tr><td>0</td><td>$\frac{9}{2}$</td></tr> <tr><td>1</td><td>$\frac{9}{4}$</td></tr> <tr><td>2</td><td>$\frac{9}{8}$</td></tr> <tr><td>3</td><td>$\frac{9}{16}$</td></tr> <tr><td>4</td><td>$\frac{9}{32}$</td></tr> </tbody> </table>	x	y	0	$\frac{9}{2}$	1	$\frac{9}{4}$	2	$\frac{9}{8}$	3	$\frac{9}{16}$	4	$\frac{9}{32}$	○ Exponential decay ○ Exponential growth
x	y												
0	$\frac{9}{2}$												
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3	$\frac{9}{16}$												
4	$\frac{9}{32}$												
Justification													



11.1.6 Wrap-Up: Reflection

1. Complete the table.

The terms and information used in the lesson that I knew were ...	Something I wondered during the lesson was ...	Something I learned during the lesson was ...

Unit 11, Lesson 2: Key Features of Exponential Functions



Benchmarks

MA.912.AR.5.3 Given a mathematical or real-world context, classify exponential functions as representing growth or decay.

MA.912.AR.5.6 Given a table, equation, or written description of an exponential function, graph that function and determine its key features.



Learning Targets

- ☐ I can determine the key features of an exponential function from a table, graph, or exponential function.
- ☐ I can classify a function as representing exponential growth or decay.



11.2.1 Warm-Up: Rhind Papyrus

Some of the earliest records of mathematical problem solving come from ancient Egypt in the form of papyri that were written between 1850 and 1600 BCE. One of these, the Rhind Mathematical Papyrus, is a collection of math problems written around 1650 BCE. The problems in the papyrus were used to manage food supplies for cities, among other things.



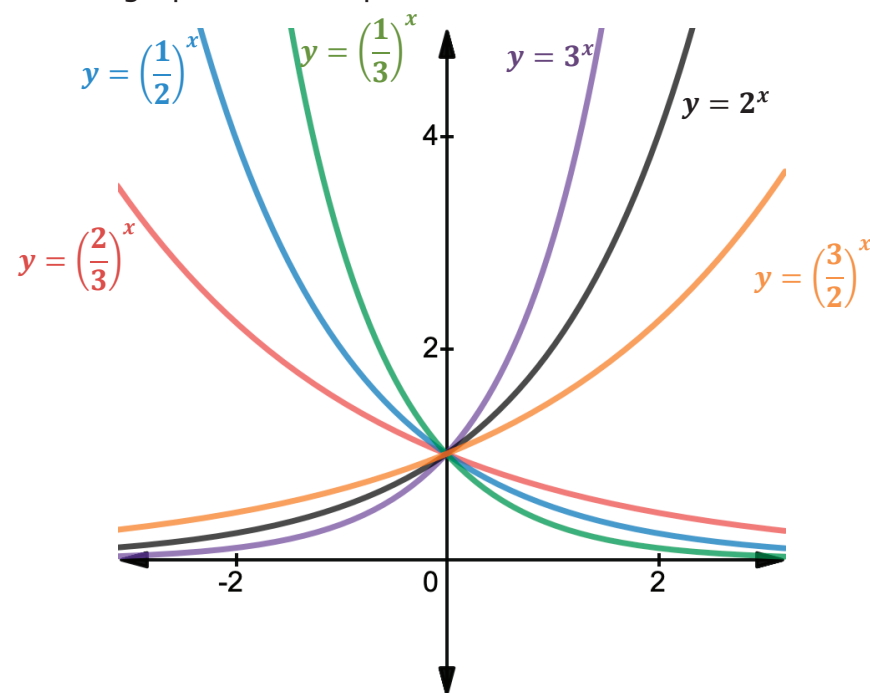
One of the problems asks: "In a certain village, there were 7 houses; each house had 7 cats; each cat caught 7 mice; each mouse would have (were it not for the cats) eaten 7 ears of spelt; each ear of spelt produced 7 hekats of grain at harvest. How many hekats of grain were saved by the presence of the cats?"

- Find the solution to the Rhind Papyrus problem.



11.2.2 Exploration: Equations, Graphs, and Tables of Exponential Functions

- The graphs of six exponential functions are shown.



- Work with your partner to complete the table.

	Graphs with a b -value Between 0 and 1	Graphs with a b -value Greater Than 1
Equations	☐ $y = \left(\frac{2}{3}\right)^x$ ☐ $y = \left(\frac{1}{2}\right)^x$ ☐ $y = \left(\frac{1}{3}\right)^x$ ☐ $y = 3^x$ ☐ $y = 2^x$ ☐ $y = \left(\frac{3}{2}\right)^x$	☐ $y = \left(\frac{2}{3}\right)^x$ ☐ $y = \left(\frac{1}{2}\right)^x$ ☐ $y = \left(\frac{1}{3}\right)^x$ ☐ $y = 3^x$ ☐ $y = 2^x$ ☐ $y = \left(\frac{3}{2}\right)^x$
Type of Function	☐ Exponential growth ☐ Exponential decay	☐ Exponential growth ☐ Exponential decay

b. Complete the statements.

For the equations that represent exponential growth, as

the x -values increase, the y -values

- ☐ increase.
- ☐ approach zero.

For the equations that represent exponential decay, as

the x -values increase the y -values

- ☐ increase.
- ☐ approach zero.

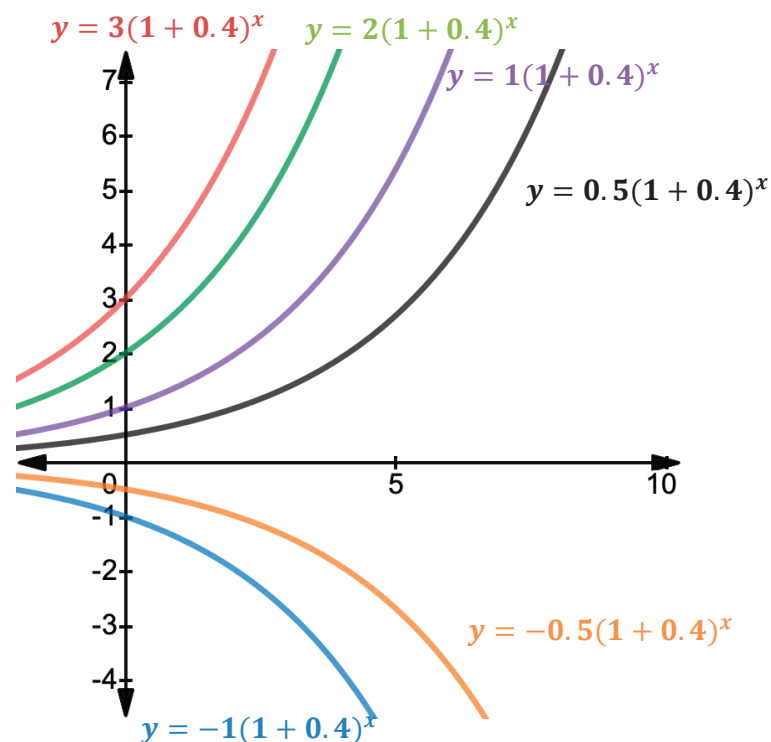
An **asymptote** is a line that a graph approaches, but never crosses or touches.

c. For the exponential decay equations, as x approaches $+\infty$, what is the horizontal asymptote?

d. For the exponential growth equations, as x approaches $-\infty$, what is the horizontal asymptote?

e. What do you notice about the graphs of exponential decay and exponential growth functions as the value of b gets closer to 1?

2. The graphs of six exponential functions are shown with their corresponding equations.



a. What do all the equations have in common?

b. What do you notice as the value of a approaches 0?

- c. Complete the table to describe the similarities and differences of the graphs of $y = -1(1 + 0.4)^x$ and $y = 1(1 + 0.4)^x$.

Similarities	Differences



11.2.3 Bring it Together: Key Features of Exponential Functions

1. Based on the 12 graphs in the Exploration, complete the table.

Key Feature	Question
Domain	Do all of the graphs have a domain of all real numbers? ☐ yes ☐ no
Range	Do all of the graphs have a range of all real numbers? ☐ yes ☐ no
x-intercepts	Do the graphs show an x -intercept? ☐ yes ☐ no
y-intercept	Do the graphs show a y -intercept? ☐ yes ☐ no
Vertex	Do the graphs show a vertex? ☐ yes ☐ no
Axis of Symmetry	Do the graphs show an axis of symmetry? ☐ yes ☐ no

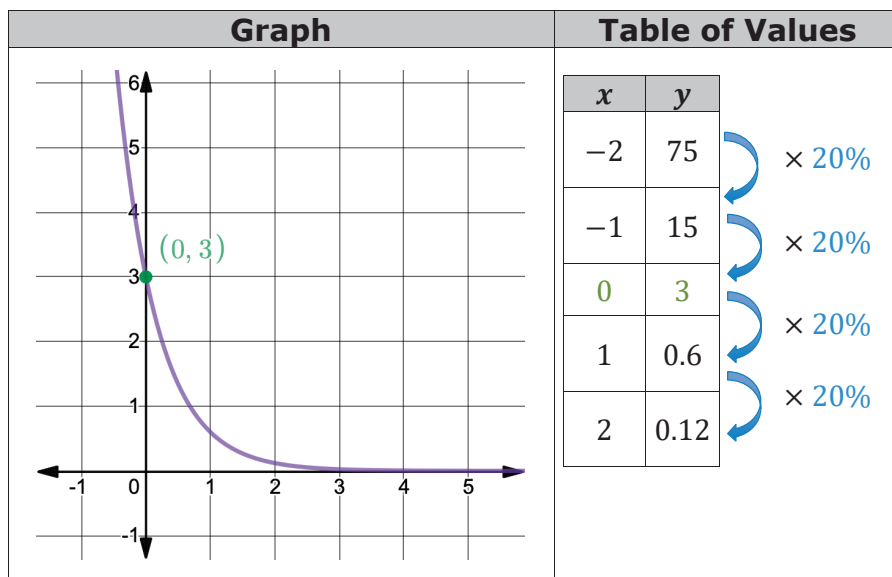
Increasing/ Decreasing	Circle the graphs that show intervals where the function is increasing.		
	$y = \left(\frac{2}{3}\right)^x$	$y = \left(\frac{1}{2}\right)^x$	$y = \left(\frac{1}{3}\right)^x$
	$y = (3)^x$	$y = (2)^x$	$y = \left(\frac{3}{2}\right)^x$
	$y = 3(1 + 0.4)^x$	$y = 2(1 + 0.4)^x$	$y = 1(1 + 0.4)^x$
	$y = 0.5(1 + 0.4)^x$	$y = -0.5(1 + 0.4)^x$	$y = -1(1 + 0.4)^x$
	Circle the graphs that show intervals where the function is decreasing.		
	$y = \left(\frac{2}{3}\right)^x$	$y = \left(\frac{1}{2}\right)^x$	$y = \left(\frac{1}{3}\right)^x$
	$y = (3)^x$	$y = (2)^x$	$y = \left(\frac{3}{2}\right)^x$
Positive/ Negative	Circle the graphs that show intervals where the function is positive.		
	$y = \left(\frac{2}{3}\right)^x$	$y = \left(\frac{1}{2}\right)^x$	$y = \left(\frac{1}{3}\right)^x$
	$y = (3)^x$	$y = (2)^x$	$y = \left(\frac{3}{2}\right)^x$
	$y = 3(1 + 0.4)^x$	$y = 2(1 + 0.4)^x$	$y = 1(1 + 0.4)^x$
	$y = 0.5(1 + 0.4)^x$	$y = -0.5(1 + 0.4)^x$	$y = -1(1 + 0.4)^x$
	Circle the graphs that show intervals where the function is negative.		
	$y = \left(\frac{2}{3}\right)^x$	$y = \left(\frac{1}{2}\right)^x$	$y = \left(\frac{1}{3}\right)^x$
	$y = (3)^x$	$y = (2)^x$	$y = \left(\frac{3}{2}\right)^x$
	$y = 3(1 + 0.4)^x$	$y = 2(1 + 0.4)^x$	$y = 1(1 + 0.4)^x$
	$y = 0.5(1 + 0.4)^x$	$y = -0.5(1 + 0.4)^x$	$y = -1(1 + 0.4)^x$

End Behavior	What do you notice about the end behavior as x -values get smaller?
	What do you notice about the end behavior as x -values get larger?
Asymptotes	What is the horizontal asymptote of exponential growth functions, $f(x) = a(b)^x$ or $f(x) = a(1 + r)^x$?
	What is the horizontal asymptote of exponential decay functions, $f(x) = a(b)^x$ or $f(x) = a(1 - r)^x$?



11.2.4 Guided Instruction: Determining Key Features of Exponential Graphs

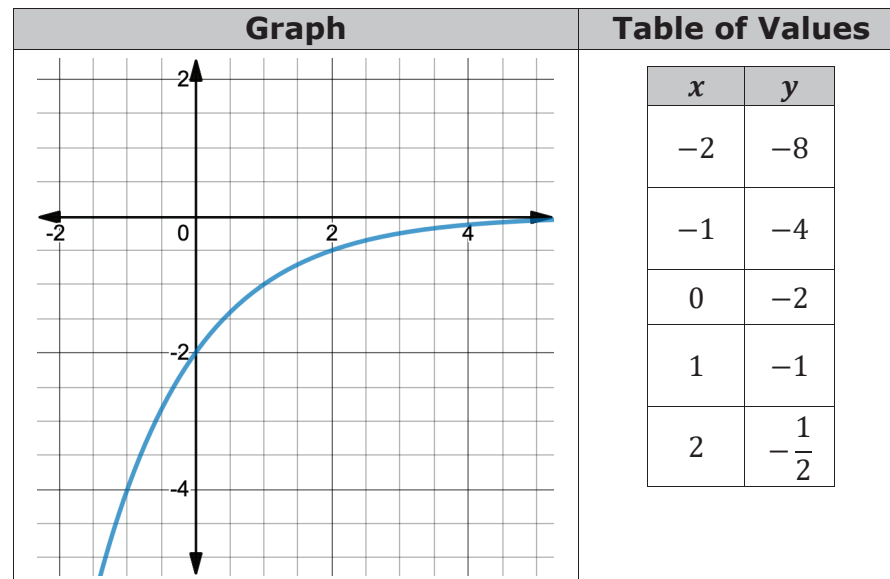
A table and graph of the exponential function $B(x) = 3(1 - 0.8)^x$ are shown. The key features of the graph are color-coded with the key features in the table.



Key Features			
Domain	$\mathbb{R}$	Range	$y > 0$
y-intercept	(0,3)	Constant Percent Rate of Change	$B(x) = 3(1 - 0.8)^x$ $B(x) = 3(0.2)^x$ $0.2 \cdot 100 = 20\%$
Increasing or Decreasing	Decreasing $-\infty < x < \infty$	Positive	$-\infty < x < \infty$
		Negative	Never negative
Left End Behavior	$x \rightarrow -\infty, y \rightarrow \infty$	Right End Behavior	$x \rightarrow \infty, y \rightarrow 0$

- Describe how the constant percent rate of change is found from the table and function.

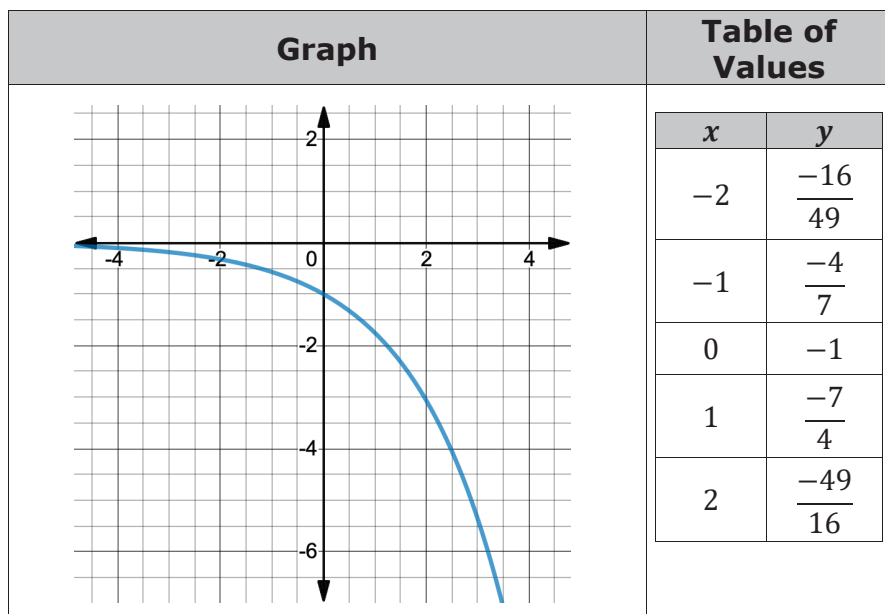
- Use the graph and table of $y = -2\left(\frac{1}{2}\right)^x$ to determine the key features.



Key Features			
Domain		Range	
y-intercept		Constant Percent Rate of Change	
Increasing or Decreasing		Positive	
		Negative	
Left End Behavior		Right End Behavior	

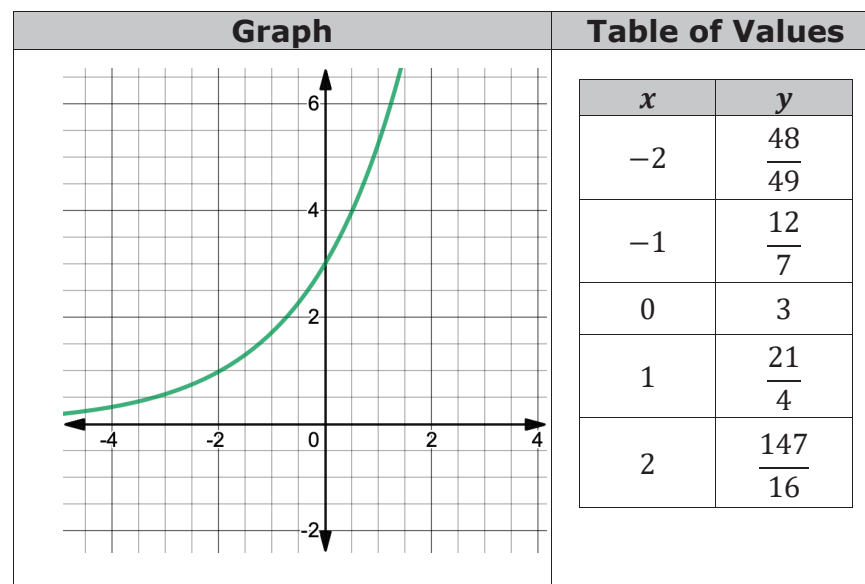


3. Use the graph and table of $y = -(1 + 0.75)^x$ to determine the key features.



Key Features			
Domain		Range	
y-intercept		Constant Percent Rate of Change	
Increasing or Decreasing		Positive	
		Negative	
Left End Behavior		Right End Behavior	

4. Use the graph and table of $B(x) = 3\left(\frac{7}{4}\right)^x$ to determine the key features.



Key Features			
Domain		Range	
y-intercept		Constant Percent Rate of Change	
Increasing or Decreasing		Positive	
		Negative	
Left End Behavior		Right End Behavior	

5. Using the key features found in the previous problems, complete the table.

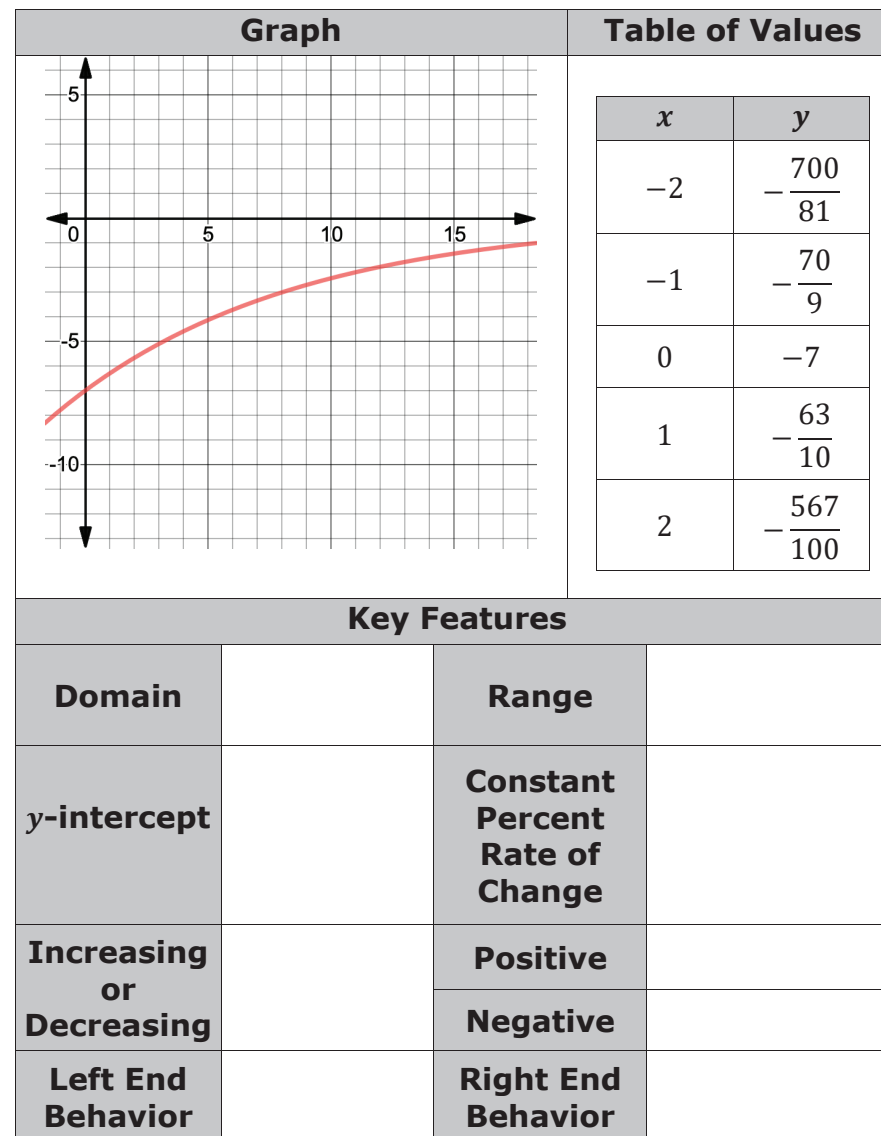
a is positive or negative	Constant Percent Rate of Change	Increasing or Decreasing	End Behavior as x Increases	End Behavior as x Decreases
Positive	Greater than 100%			
Positive	Between 0% and 100%			
Negative	Greater than 100%			
Negative	Between 0% and 100%			

6. Discuss with your partner what effect the a -value and the constant percent rate of change have on the graph. Write a summary of your discussion.



11.2.5 Your Turn!

1. Use the graph and table of $y = -7(1 - 0.1)^x$ to determine the key features.





11.2.6 Wrap-Up: Reflection

1. Circle the icon that best describes your understanding of each key feature.



I need help.



I got it.



I could help someone.

Finding domain and range of an exponential function			
Finding the constant percent rate of change			
Describing the end behavior of an exponential function			
Finding the y -intercept			
Finding the intervals where the function is positive or negative			
Finding the intervals where the function is increasing or decreasing			
Identifying the horizontal asymptote			

Unit 11, Lesson 3: Graphing Exponential Functions Given an Equation



Benchmarks

MA.912.AR.5.3 Given a mathematical or real-world context, classify exponential functions as representing growth or decay.

MA.912.AR.5.6 Given a table, equation, or written description of an exponential function, graph that function and determine its key features.



Learning Targets

- ☐ I can graph an exponential function given the equation.
- ☐ I can classify a function as representing exponential growth or decay.



11.3.1 Warm-Up: Registered Nurse

1. Keith Koga is a registered nurse. He said he's had a lot of mentors, some of which included nurses he worked with. The culture of the workplace was one where everyone supported each other and knew they could go to each other for help.
 - a. Who is someone you consider a mentor in your life?
 - b. When learning something new, why is it helpful to ask someone (a peer, a teacher, a mentor, etc.) when you need help?



11.3.2 Guided Instruction: Graphing an Exponential Equation

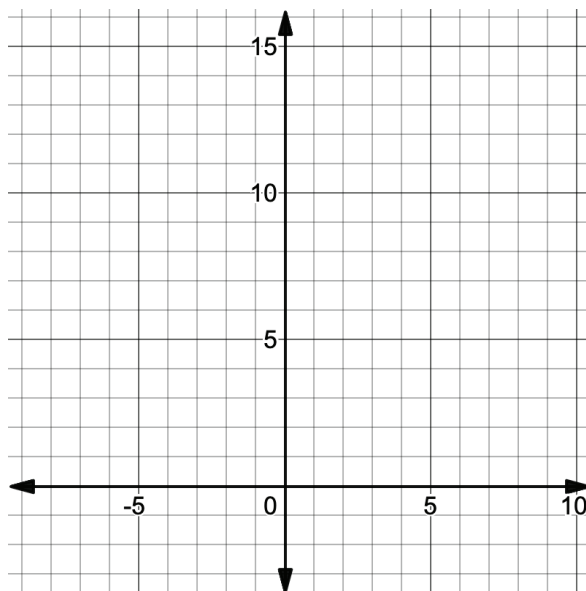
When graphing an exponential function, knowing the key features and using a table of values can be helpful.

1. Explain what key features can be determined from the equation $C(x) = 5\left(\frac{6}{5}\right)^x$.

Key Feature	Explanation
Growth or Decay	
Domain and Range	
Increasing / Decreasing	
Constant Percent Rate of Change	
End Behavior	
Positive / Negative	

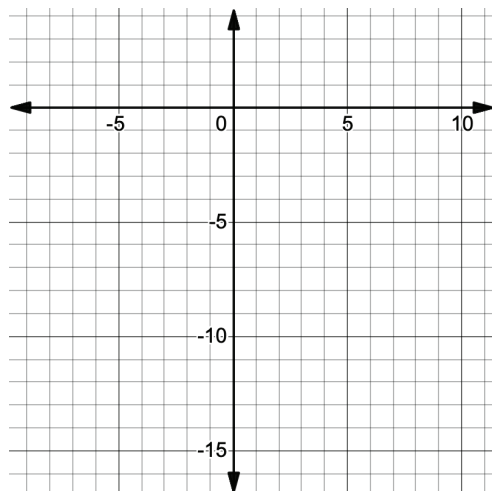
2. Complete the table for $C(x) = 5\left(\frac{6}{5}\right)^x$. Then, graph the function.

x	$C(x)$
-2	
-1	
0	
1	
2	
3	



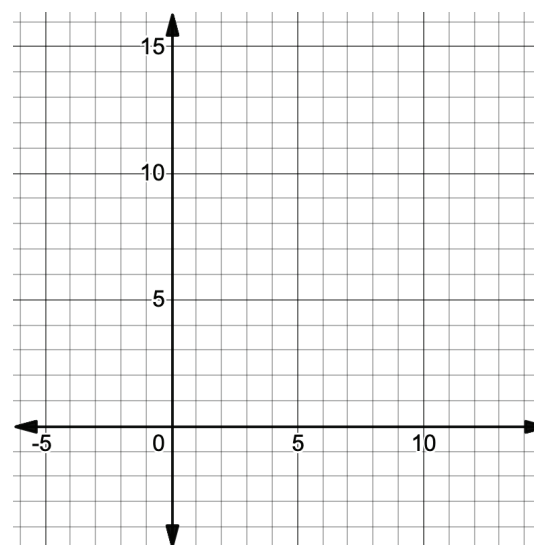
- a. Discuss with your partner how the graph would change if the a -value of 5 was changed to -5 .

- b. Draw a sketch of $C(x) = -5\left(\frac{6}{5}\right)^x$.



- c. Discuss with your partner how the graph would change if the b -value was changed from $\frac{6}{5}$ to $\frac{5}{6}$.

- d. Draw a sketch of $C(x) = 5\left(\frac{5}{6}\right)^x$.





11.3.3 Collaboration: Graphing Exponential Functions

1. Consider the function $f(x) = 3(1 - 0.32)^x$.

a. Work with your partner to complete the statements about $f(x) = 3(1 - 0.32)^x$.

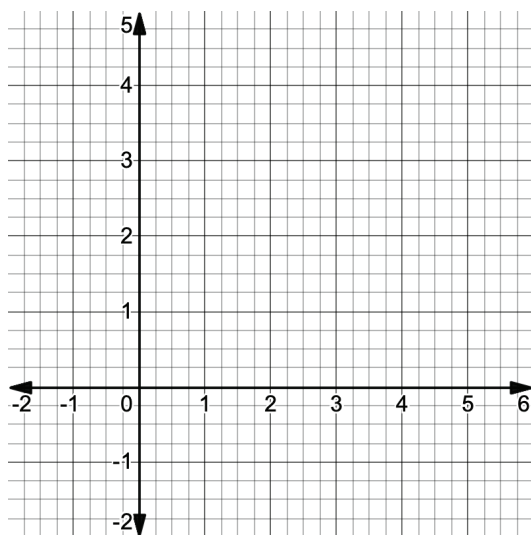
Since a is ☐ positive ☐ negative and b is ☐ between 0 and 1, ☐ greater than 1,

the function is ☐ increasing. ☐ decreasing. The left end behavior of

the graph is ☐ $y \rightarrow 0$ ☐ $y \rightarrow \infty$ and the right end behavior is

☐ $y \rightarrow 0$. ☐ $y \rightarrow -\infty$. The y -intercept of the function is ☐ 0.32. ☐ 0.68. ☐ 3.

b. Sketch the graph of $f(x) = 3(1 - 0.32)^x$.



2. Consider the function $f(x) = -(1 + 0.4)^x$.

a. Work with your partner to complete the statements for $f(x) = -(1 + 0.4)^x$.

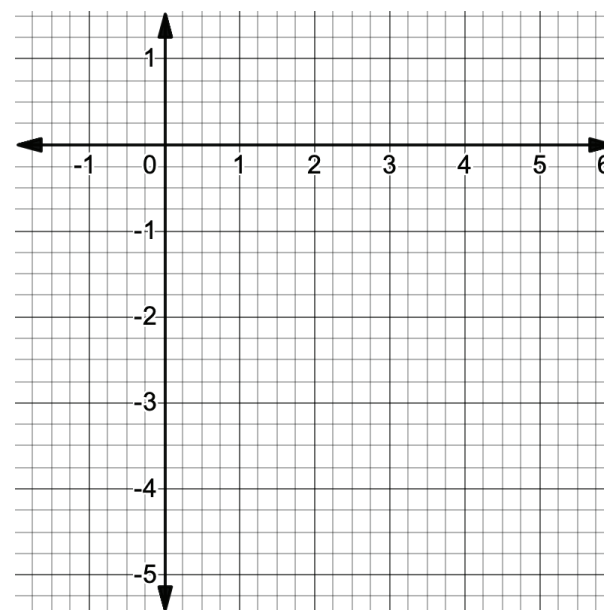
Since a is ☐ positive ☐ negative and b is ☐ between 0 and 1, ☐ greater than 1,

the function is ☐ increasing. ☐ decreasing. The left end behavior of

the graph is ☐ $y \rightarrow 0$ ☐ $y \rightarrow -\infty$ and the right end behavior is

☐ $y \rightarrow 0$. ☐ $y \rightarrow -\infty$. The y -intercept of the function is ☐ 0.4. ☐ -1. ☐ 1.04.

b. Sketch the graph of $f(x) = -(1 + 0.4)^x$.





11.3.4 Activity: Card Sort

Work with your partner to complete the card sort. Fill in the table with the correct card labels that correspond to each graph.

Card Sort					
Graph	y-Intercept	$a =$ $b =$	Function	Table	Table
1.1					
1.2					
1.3					
1.4					
1.5					
1.6					



11.3.5 Your Turn!

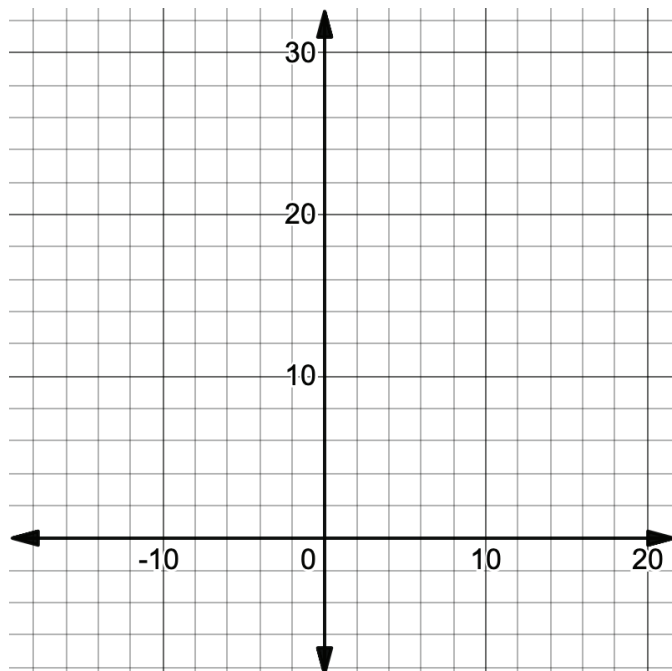
- Graph the exponential equation $y = 45(1 - 0.05)^x$. If necessary, complete the table of values. Then, determine all the key features.

Graph		Table of Values	
		x	y
		-2	
		-1	
		0	
		1	
		2	
Key Features			
Domain		Range	
y-intercept		Constant Percent Rate of Change	
Increasing or Decreasing		Positive	
		Negative	
Left End Behavior		Right End Behavior	



11.3.6 Wrap-Up: Ticket Out the Door

1. Graph the function $y = 9\left(\frac{5}{3}\right)^x$.



Unit 11, Lesson 4: Graphing Exponential Functions Given a Written Description and Equation



Benchmarks

MA.912.AR.5.3 Given a mathematical or real-world context, classify exponential functions as representing growth or decay.

MA.912.AR.5.6 Given a table, equation, or written description of an exponential function, graph that function and determine its key features.



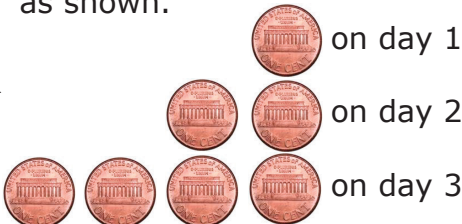
Learning Targets

- ☐ I can graph an exponential function given a written description and equation.
- ☐ I can determine the key features of an exponential function.
- ☐ I can classify a function as representing exponential growth or decay.



11.4.1 Warm-Up: Would You Rather?

1. Imagine you have the chance to pick from one of the two options shown.

Option A	Option B
Receive a one-time payment of \$1 million dollars.	Receive an increasing number of pennies each day as shown.
	
	OR
	and so on for 1 month

- a. Would you rather choose Option A or Option B?
- b. Using mathematics, show which option pays out more money.



11.4.2 Guided Instruction: Graphing Exponential Functions from a Written Description and Equation

The half-life of a drug is an estimated period of time that it takes for an amount of that drug in the body to be reduced by exactly half. The amount, A , in milligrams, of a drug in the body for h hours after taking a pill is given by

$$A(h) = 50 \left(\frac{1}{2}\right)^h.$$

1. Graph the function that models the situation.



2. Describe how to determine whether the situation models exponential growth or decay.

3. What is the initial dose given?

4. Describe how to identify the initial dose from the function and from the graph in the table.

From the Function	From the Graph

5. What percentage of the drug leaves the body each hour?

6. How is this percentage represented in the equation?

7. Complete the table to identify other key features of the function based on the context. In the second row, explain how each key feature relates to the context.

	Domain	Range	As x -values increase, y -values ...
Key Feature			
Reason for Constraint from Context			



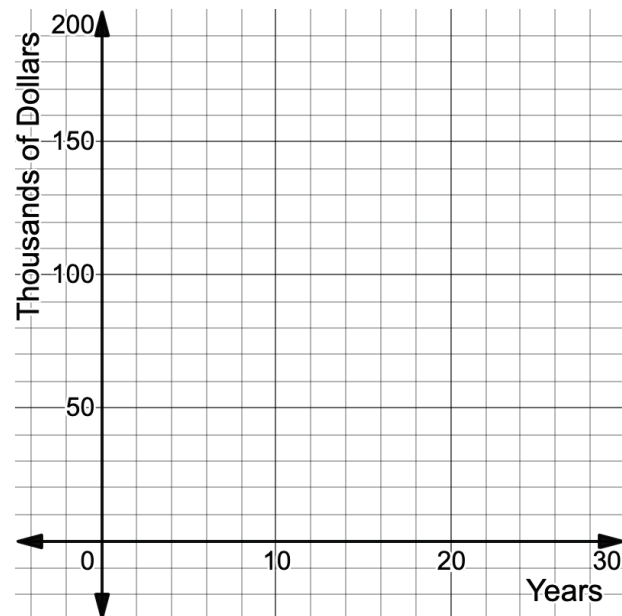
11.4.3 Collaboration: Graphing Exponential Functions

Liam recently graduated from the University of South Florida and accepted a job as a software engineer. The starting salary is \$50,000 and has a guaranteed annual pay increase of 3% each year.

The equation $y = 50,000(1 + 0.03)^x$ models the increase in salary over time, where x is the number of years since he started at the company and y is his salary.

1. Describe how to determine if this situation models exponential growth or decay.

2. Graph the equation $y = 50,000(1 + 0.03)^x$.



3. What does the y -intercept represent in this situation?

4. Harper said that the domain is $0 \leq x \leq \infty$. Discuss with your partner if Harper is correct. Write a summary of your discussion.

5. Describe why the range depends on the number of years the person works for the company.

6. Complete the statements.

If Liam works for the company for 30 years, the domain is _____ $\leq x \leq$ _____, with a range of _____ $\leq y \leq$ _____.

7. Two students were asked to find the constant percent rate of change for this situation. Their answers are shown.

Owen	Fiona
The constant percent rate of change for the situation is 3% because Liam receives a pay increase of 3% every year.	The constant percent rate of change is 103% because each year Liam receives a 3% pay increase in addition to 100% of the salary he earned the year before.

- Discuss with your partner who identified the correct constant percent rate.
- Explain the mistake made by the student who was incorrect.



11.4.4 Activity: Gallery Walk

- Work with a partner to graph the situation below that your teacher has chosen for you.

Situation A	Situation B
The annual rate of inflation for goods and services is modeled by $C(t) = P(1.024)^t$, where C is the approximate cost of goods or services, t is the time in years, and P is the present cost. If the present cost of Nike LeBron basketball shoes is \$149.99, graph the function that will model the cost of shoes over time.	A new 2021 Ford Ranger XLT costs \$31,045 with a depreciation rate of 8% per year. This situation is modeled by the function $V(x) = 31,045(1 - 0.08)^x$, where x is the number of years since 2021 and V is the value of the car.
Situation C	Situation D
One hundred bacteria cells are put into a petri dish, and the culture doubles in size every hour. This is modeled by $P(h) = 100(2)^h$, where P is the number of bacteria in the population and h is the number hours that have passed.	A new computer that was recently purchased depreciates over time by the given function, $C(x) = \$1,200\left(\frac{1}{4}\right)^x$, where x is the number of years since the computer was purchased and C is the current value of the computer.

2. Once you and your partner graph the function, determine the key features and interpret how each relates to the context.

	Key Feature	Interpretations
Domain		
Range		
y-intercept		
Constant Percent Rate of Change		
End Behavior		



11.4.5 Wrap-Up: Growth Mindset

Which mindset did I demonstrate?	Choose the best description.
Did I use the partner discussions as learning opportunities?	☐ Never ☐ Sometimes ☐ Usually ☐ Always
Did I work on problems that challenged me?	☐ Never ☐ Sometimes ☐ Usually ☐ Always
Did I use strategies to 'un-stick' myself when I found the problems difficult?	☐ Never ☐ Sometimes ☐ Usually ☐ Always
Did I put as much effort as I possibly could into the problems?	☐ Never ☐ Sometimes ☐ Usually ☐ Always

Unit 11, Lesson 5: Writing Exponential Functions from a Graph or Table



Benchmark

MA.912.AR.5.4 Write an exponential function to represent a relationship between two quantities from a graph, a written description, or a table of values within a mathematical or real-world context.



Learning Targets

- ☐ I can write an exponential function from a graph.
- ☐ I can write an exponential function from a table.



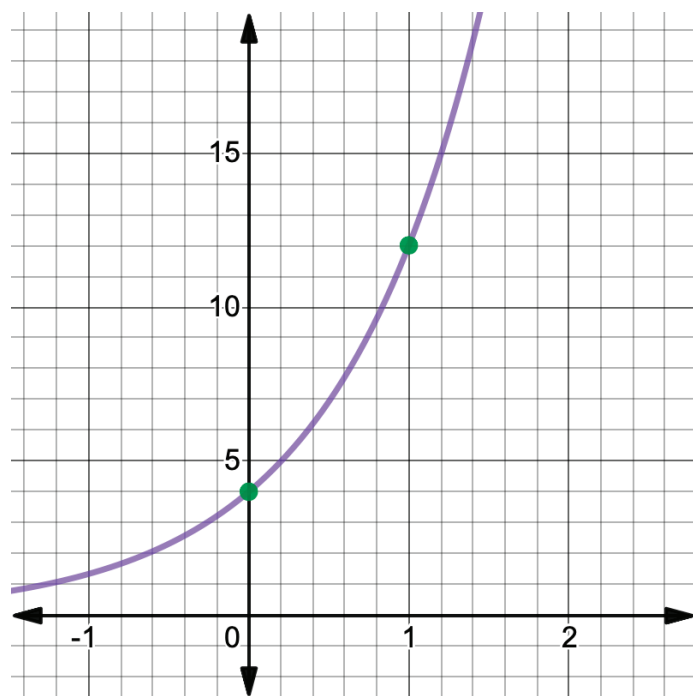
11.5.1 Warm-Up: Growth Mindset

1. It's crucial to create an environment where failure, errors, and mistakes are seen as a natural part of learning. Making that environment all starts with you; more specifically, your self-talk.
 - a. When you make a mistake, what do you say to yourself?
 - b. Explain if what you say to yourself reflects a growth or fixed mindset.



11.5.2 Guided Instruction: Writing an Exponential Function from Graphs and Tables

1. The graph of an exponential function is shown.



To express this exponential function in the form $f(x) = a(b)^x$ or $f(x) = a(1 \pm r)^x$, the values for a and b need to be found. These values can be found using two points on the graph and the equation in the form $y = a(b)^x$.

- a. Find the value of a by substituting point $(0, 4)$ in for x and y in $y = a(b)^x$.

$$\boxed{} = a(b)^{\boxed{}}$$

- b. Discuss with your partner if the point $(1, 12)$ can be used to determine the value of b . Write a brief summary of your discussion.

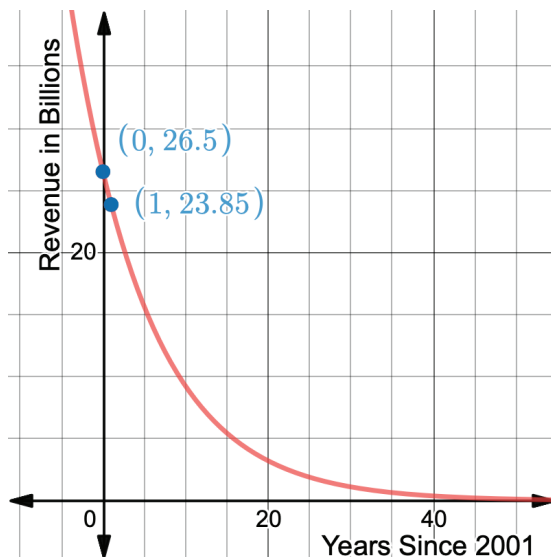
- c. Use the value of a found in Part A and the point $(1, 12)$ to determine the value of b .

$$\boxed{} = \boxed{}(b)^{\boxed{}}$$

- d. Write the function for the graph using both forms.

$f(x) = a(b)^x$	$f(x) = a(1 \pm r)^x$

2. The Global Music Report shows there has been a decline in revenue from physical records due to the increase of digital downloads and streaming services. The graph shows y , the revenue in billions, and x , years since 2001.



- a. Discuss with your partner which point can be used to find the a value for the equation $y = a(b)^x$.
- b. Work with your partner to write the equation for the revenue of physical records since 2001.

- c. What does the a value represent in the context?

- d. What does the b value represent in the context?

3. Writing an equation from a table of values can be done by finding the initial value, a , and the common ratio, b .

The table of values represents an exponential relationship.

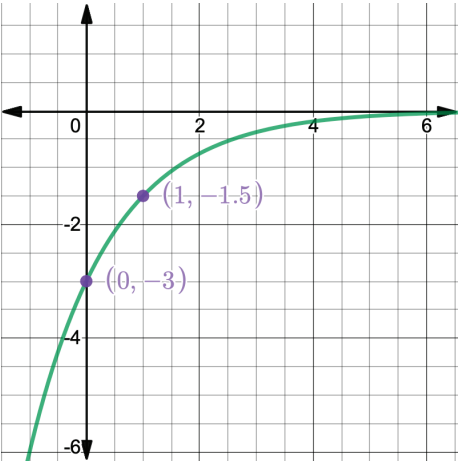
x	0	1	2	3	4	5	6
$f(x)$	$\frac{2}{3}$	2	6	18	54	162	486



- a. Describe how the values of $f(x)$ change as the x -values increase by 1.
- b. What is the initial value, a ?

- c. Discuss with your partner how to use the information determined in Parts A and B to write the function in the form $f(x) = a(b)^x$. Then, write the function.

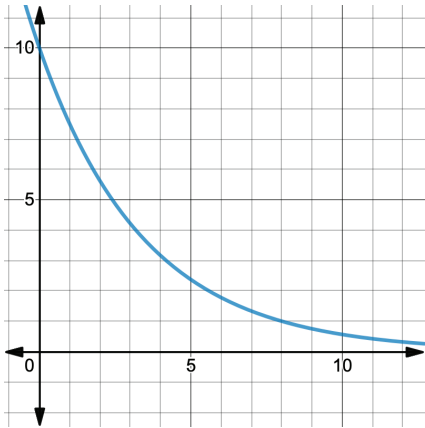
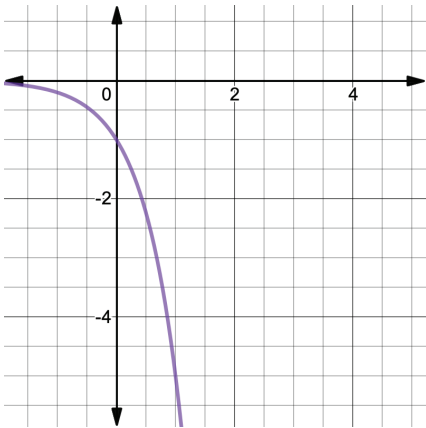
4. When the value of a is negative, the same process can be used. Write an exponential function for the graph and for the table of values in the form $f(x) = a(b)^x$.

Graph	Table												
	<table> <tr> <th>x</th><th>$f(x)$</th></tr> <tr> <td>0</td><td>-2</td></tr> <tr> <td>1</td><td>-7</td></tr> <tr> <td>2</td><td>-24.5</td></tr> <tr> <td>3</td><td>-85.75</td></tr> <tr> <td>4</td><td>-300.125</td></tr> </table>	x	$f(x)$	0	-2	1	-7	2	-24.5	3	-85.75	4	-300.125
x	$f(x)$												
0	-2												
1	-7												
2	-24.5												
3	-85.75												
4	-300.125												
Function	Function												



11.5.3 Guided Practice: Writing Exponential Functions

1. Write the exponential function for each graph.

	
Function	Function

2. Write an exponential function for each table of values.

x	$f(x)$
0	4
1	10
2	25
3	62.5
4	156.25

x	$f(x)$
0	20
1	16
2	$\frac{64}{5}$
3	$\frac{256}{25}$
4	$\frac{1,024}{125}$

Function

Function



11.5.4 Wrap-Up: Ticket Out the Door

1. Write the exponential function represented by the graph and the exponential function represented in the table of values.

Graph	Table												
	<table> <tr> <th>x</th><th>$f(x)$</th></tr> <tr> <td>0</td><td>0.7</td></tr> <tr> <td>1</td><td>3.5</td></tr> <tr> <td>2</td><td>17.5</td></tr> <tr> <td>3</td><td>87.5</td></tr> <tr> <td>4</td><td>437.5</td></tr> </table>	x	$f(x)$	0	0.7	1	3.5	2	17.5	3	87.5	4	437.5
x	$f(x)$												
0	0.7												
1	3.5												
2	17.5												
3	87.5												
4	437.5												
Function	Function												
$f(x) =$	$f(x) =$												

Unit 11, Lesson 6: Writing Exponential Functions from a Written Description



Benchmarks

MA.912.AR.5.3 Given a mathematical or real-world context, classify exponential functions as representing growth or decay.

MA.912.AR.5.4 Write an exponential function to represent a relationship between two quantities from a graph, a written description, or a table of values within a mathematical or real-world context.



Learning Targets

- ☐ I can write an exponential function from a written description.
- ☐ I can classify a real-world context modeled by an exponential function as exponential growth or decay.



11.6.1 Warm-Up: Math Talk

The map shows the Southeastern United States to scale. Using the map, fill in the blanks below. Be prepared to share your reasoning with the class.

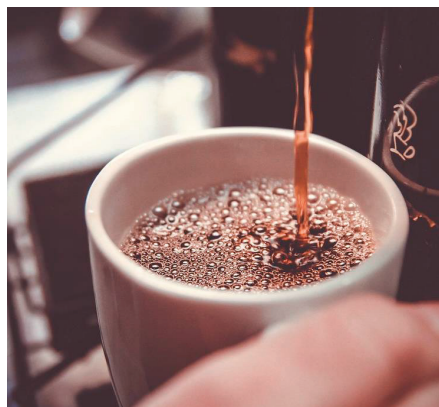


1. Florida has an area that is about _____ times bigger than South Carolina.
2. The area of Florida is 65,755 square miles. Based on your estimate, the area of South Carolina is _____ square miles.



11.6.2 Guided Instruction: Writing Exponential Functions

1. A typical cup of coffee contains about 100 milligrams (mg) of caffeine. Every hour approximately 6% of the amount of caffeine in the body is metabolized and eliminated.



- a. Discuss with your partner whether the situation represents exponential growth or exponential decay. Justify your choice.
- b. What is the rate of growth or the rate of decay?
- c. What is the initial amount of caffeine consumed in this scenario?
- d. Let C represent the amount of caffeine in the body, in mg, and t represent the number of hours since a cup of coffee was consumed. Write C as a function of t .

2. In 2020, the population of Florida was estimated to be 21.7 million people, with an annual rate of increase of 1.12%.

- a. Explain whether the population can be modeled by exponential growth or exponential decay.
- b. Wyatt and Carter both wrote a function to model Florida's population growth, where x is the number of years since 2020 and P is the population in millions. Both students made a mistake. Circle the error each student made.

Wyatt	Carter
$P(x) = 21.7(1.12)^x$	$P(x) = 21.7(1 + 0.112)^x$

- c. Discuss with your partner the errors Wyatt and Carter made.
- d. Write the correct function that models the context using both forms.

$P(x) = a(b)^x$	$P(x) = a(1 + r)^x$



11.6.3 Try It: Writing Exponential Functions

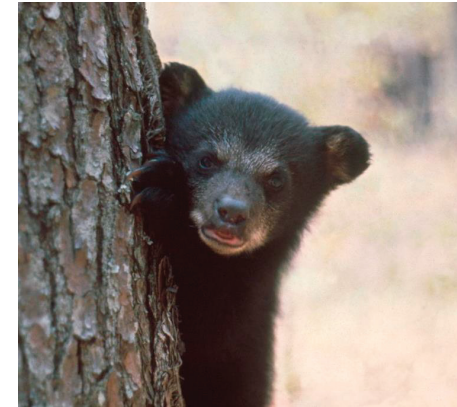
Work independently to write the function for the two scenarios.

1. The value of a 2021 Ford Bronco was \$44,205 when it went on the market. It depreciates at a rate of 10.5% per year.



- a. Is this an example of growth or decay?
- b. Write a function that models the car's value, where V is the value of car and t is the number of years since 2021.

2. The Florida Fish and Wildlife Conservation Commission have found that the Florida black bear population is increasing due to the Florida Black Bear Management Plan. In 2019, there were 4,050 bears in Florida with a rate of increase of 6.5% per year.



- a. Is this an example of growth or decay?
- b. Write an equation that models the population, where B is the number of black bears and x is the years since 2019.

3. Compare your answers to questions 1-2 with your partner and if necessary, resolve any differences.



11.6.4 Collaboration: Exponential Equations in Real-World Contexts

U.S. lawmakers have tried to pass legislation that would reduce the consumption of salt. Versions of this law have been proposed in the past that would mandate restaurants to decrease sodium levels in the food they serve by an exponential rate. The sodium level in food is the amount of salt in the food.

It is estimated that all U.S. restaurants use a collective total of 5 million grams of salt a year. If the law were to pass, they would be required to reduce to 4.875 million grams the following year and continue reducing at a constant percentage each year until a goal is reached.

1. Discuss with your partner how the values given in this scenario can be used to write an exponential function that models the situation.
2. Two students discussed how to determine the value of b to write an exponential function in the form $y = a(b)^x$. Their thinking is shown in the table.

Jayden	Ella
"I would subtract 4.875 million from 5 million to find the rate of change from one year to the next."	"I would divide 4.875 million by 5 million to find the common ratio from one year to the next."

Explain which student is correct.

3. Find the value of b in this scenario.
4. Discuss with your partner which value in the scenario represents a .
5. Work with your partner to write the function rules in the forms indicated, where S is the total amount of salt that can be used, in millions of grams, for the given years, x , since the law passed.
6. What is the constant percent rate of change by which the law would require restaurants to reduce the amount of salt served in their food each year?

$S(x) = a(b)^x$	$S(x) = a(1 \pm r)^x$



11.6.5 Your Turn!

1. In 1987, there were 17 Starbucks locations. The number of shops has increased 29% annually since then.

Write the function that models the number of Starbucks since 1987. Let t be the number of years since 1987 and S be the number of shops.

2. There were 1.462 million paid subscriptions for streaming music in 2010. Since then, there has been a 54.4% per year increase in paid music streaming subscriptions.

Write a function that represents the number of paid subscriptions since 2010. Let x be the years since 2010 and P be the number of paid music streaming subscriptions, in millions.

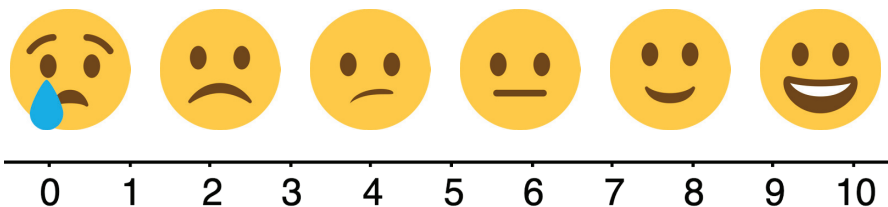
3. An adult takes 400 milligrams (mg) of ibuprofen. Each hour, the amount of ibuprofen in the person's system decreases by 29%.

Write a function that represents the amount of ibuprofen, A , in mg, after h hours.



11.6.6 Wrap-Up: Reflection

A scale is shown with emojis that correlate to different levels of confidence.



1. Draw the emoji that best describes your understanding of each learning target from this unit.

Emoji	Learning Target
	I can classify a function as representing exponential growth or decay from a graph or table.
	I can determine the key features of an exponential function from a table, graph, or exponential function.
	I can graph an exponential function given the equation.
	I can graph an exponential function given a written description and determine its key features.
	I can write an exponential function from a graph or table.
	I can write an exponential function from a written description.

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